FN FAL / L1A1 SLR Battle Rifle

FN Herstal / Royal Small Arms Factory Enfield / Imbel / DSA • 'The Right Arm of the Free World' (7.62x51mm NATO Metric & Inch Pattern)

Manufacturer:	FN Herstal / Royal Small Arms Factory Enfield / Imbel / DSA	**Category:**	Rifles & Carbines
Action Type:	Short-Stroke Gas Piston Tilting-Breechblock Battle Rifle	**Era / Service:**	Cold War Free World Standard (1953–Present)
Calibers:	7.62x51mm NATO, .308 Winchester	**Covered Models:**	FAL 50.00 (Standard), FAL 50.63 (Para Folding Stock), L1A1 SLR (British Commonwealth Inch), StG 58 (Austrian)

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Locking Shoulder Sizing (Pin Gauge Headspacing) • Gas Regulator Vent Sleeve Calibration (1–7 clicks) • Hinge Pin Takedown & Break-Open Action • Tilting Bolt Lug Lapping • Recoil Spring Tube Servicing

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
HEADSPACE METHOD	Headspace is set by pressing a precision-ground cylindrical locking shoulder (available in 0.001" increments from 0.254" to 0.272") into the receiver floor
GAS REGULATOR CALIBRATION	Gas sleeve vents excess gas. Standard armorer calibration: Open regulator wide until rifle fails to lock open on empty magazine, then close regulator sleeve 2 clicks for 100% reliable combat operation
BOLT LOCKING	Rear of breechblock drops downward into locking shoulder recess in floor of receiver
METRIC VS INCH	Metric magazines have small front beak; Inch magazines have large heavy brazed lug (Inch receivers can use Metric mags, but not vice-versa)

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Violent Over-Gassed Ejection / Blown Case Heads Rifle ejects brass with extreme violence, tearing extractor rim or kicking shooter hard	• Gas regulator sleeve screwed shut (vent hole completely covered). • Shooting hot commercial ammunition with military gas setting.	**Immediate Action: Rotate gas regulator collar clockwise (numbers getting higher) to open vent port and release excess gas until ejection drops to a comfortable 10–12 feet.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
FN FAL Complete Armorer Headspace Setting & Locking Shoulder Pin Gauging	Mark Novak / Anvil Gunsmithing	Step-by-step masterclass on setting headspace on the FN FAL using Clymer gauges, pin gauge rods, selecting locking shoulders, and tuning the gas collar.